

Morning Paper Report - Friday, September 25, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Recurrent neural networks with transient trajectory explain working memory encoding mechanisms

Chenghao Liu, Shuncheng Jia, Hongxing Liu, Xuan-Le Zhao, Chengyu T Li, Bo Xu, et al.
Communications Biology, 2025

Source: Semantic Scholar; matched terms: working memory, modeling, neural network, neural networks, recurrent neural network

Link: <https://www.semanticscholar.org/paper/4e54136ec70defde781d3473203eb1d0c9083035>

Goal: Whether working memory (WM) is encoded by persistent activity using attractors or by dynamic activity using transient trajectories has been debated for decades in both experimental and modeling studies, and a consensus has not been reached

Method: Whether working memory (WM) is encoded by persistent activity using attractors or by dynamic activity using transient trajectories has been debated for decades in both experimental and modeling studies, and a consensus has not been reached

Results: Even though many recurrent neural networks (RNNs) have been proposed to simulate WM, most networks are designed to match respective experimental observations and show either transient or persistent activities

Conclusion: Therefore, this study provides evidence supporting the transient activity theory to explain the WM mechanism from the model designing point of view

2. Semantic meaning enhances feature-binding but not quantity or precision of locations in visual working memory

Tomer Sahar, N. Gronau, T. Makovski
Memory & Cognition, 2024

Source: Semantic Scholar; matched terms: visual working memory, working memory, precision, modeling

Link: <https://www.semanticscholar.org/paper/9cdadcfa35b7756a50763aa4cf5c5c691b7bf60b>

Goal: Recent studies showed that real-world items are better remembered in visual working memory (VWM) than visually similar stimuli that are stripped of their semantic meaning

Method: Employing a mixture-modeling analysis, we examined whether semantic meaning enables more item locations to be remembered, whether it improves the precision of the locations stored in memory, or whether it improves binding between the specific items and their locations

Results: Recent studies showed that real-world items are better remembered in visual working memory (VWM) than visually similar stimuli that are stripped of their semantic meaning

Conclusion: In line with previous findings, these results indicate that conceptual meaning enhances VWM for arbitrary stimulus properties such as item location, and this improvement is primarily due to a more efficient identity-location binding rather than an increase in the quantity or quality (precision) of the locations held in memory.

3. Random noise promotes slow heterogeneous synaptic dynamics important for robust working memory computation

Nuttida Rungratsameetaweemana, Robert Kim, Thiparat Chotibut, T. Sejnowski

Proceedings of the National Academy of Sciences of the United States of America, 2025

Source: Semantic Scholar; matched terms: working memory, modeling, neural network, neural networks, recurrent neural network

Link: <https://www.semanticscholar.org/paper/67684ed80ce285d11fbfe34e1384b9d494db2413>

Goal: Significance Our study addresses a fundamental challenge in the field of neural network modeling, offering critical insights into the training of recurrent neural networks (RNNs) that simulate cognitive processes

Method: Significance Our study addresses a fundamental challenge in the field of neural network modeling, offering critical insights into the training of recurrent neural networks (RNNs) that simulate cognitive processes

Results: Further analyses revealed that the introduction of noise selectively increased the synaptic decay time constants of inhibitory units, leading to a sustained stimulus-specific activity crucial for memory maintenance

Conclusion: Further analyses revealed that the introduction of noise selectively increased the synaptic decay time constants of inhibitory units, leading to a sustained stimulus-specific activity crucial for memory maintenance Our findings not only shed light on the optimization of RNN training methods but also hold profound implications for understanding how higher cortical areas evolved to compensate for inherent noise to...

4. Myelin dystrophy impairs signal transmission and working memory in a multiscale model of the aging prefrontal cortex

Sara Ibañez, Nilapratim Sengupta, J. Luebke, Klaus Wimmer, C. M. Weaver

eLife, 2024

Source: Semantic Scholar; matched terms: working memory, computational model, modeling, neural network

Link: <https://www.semanticscholar.org/paper/16d711262816f8ce9156387bd977287f5ccaad26>

Goal: Normal aging leads to myelin alterations in the rhesus monkey dorsolateral prefrontal cortex (dlPFC), which are positively correlated with degree of cognitive impairment

Method: It is hypothesized that remyelination with shorter and thinner myelin sheaths partially compensates for myelin degradation, but computational modeling has not yet explored these two phenomena together systematically

Results: Next, we incorporated the single neuron results into a spiking neural network model of working memory

Conclusion: While complete remyelination nearly recovered axonal transmission and network function to unperturbed levels, our models predict that biologically plausible levels of myelin dystrophy, if uncompensated by other factors, can account for substantial working memory impairment with aging The present computational study unites empirical data from ultrastructure up to behavior during normal aging, and has broader...

5. The objecthood of visual working memory representations influences their interaction with attention.

Chen L, Zheng J, Xu M, Zhu P, Fu Y, Shen M, et al.

Cognition, 2026

Source: Europe PMC; matched terms: visual working memory, working memory representation, working memory

Link: <https://doi.org/10.1016/j.cognition.2026.106727>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

